EMS612U Flight dynamics and control
Tutorial 3 typed solutions

3.1

$$\eta_{trim}=\frac{C_{m0}}{V_T a_2}-\frac{a_1\eta_T}{a_2}+\frac{C_L}{\bar{V}_T a_2}\left[(h-h_0)-\bar{V}_T\frac{a_1}{a}\left(1-\frac{d\varepsilon}{d\alpha}\right)\right]$$

$$\eta_T=\frac{a_2}{a_1}\left\{\frac{C_{m0}}{\bar{V}_T a_2}-\eta_{trim}+\frac{C_L}{\bar{V}_T a_2}\left[(h-h_0)-\bar{V}_T\frac{a_1}{a}\left(1-\frac{d\varepsilon}{d\alpha}\right)\right]\right\}$$

$$\eta_T=\frac{1.2}{2.8}\left\{\frac{-0.016}{0.48\cdot 1.2}-0+\frac{0.3}{0.48\cdot 1.2}\left[0.09-0.48\frac{1.8}{4.5}(1-0.4)\right]\right\}=-0.0318\ \text{rad}=-1.82^0$$

$$h=h_0+\bar{V}_T\frac{a_1}{a}\left(1-\frac{d\varepsilon}{d\alpha}\right)+\frac{\bar{V}_T a_2}{C_L}\left[\eta_{trim}-\frac{C_{m0}}{\bar{V}_T a_2}+\frac{a_1\eta_T}{a_2}\right]$$

$$h=0.18+\frac{0.48\cdot 2.8\cdot 0.6}{4.5}+\frac{0.48\cdot 1.2}{1.4}\left(\frac{-30\cdot\pi}{180}+\frac{0.016}{0.48\cdot 1.2}-\frac{2.8\cdot 0.0318}{1.2}\right)=0.125$$

3.2

(i)

$C_m=C_{m0}+(h-h_0)C_L-\bar{V}_T C_L^T$ (from lecture, pitching moment equation)

$$C_m=0$$

$$C_L=-\frac{C_{m0}}{h-h_0-\bar{V}_T C_L^T/C_L}$$

$$\bar{V}_T=\frac{S_T l_T}{S\bar{\bar{c}}}=\frac{5\cdot 10}{40\cdot 2}=\frac{5}{8}$$

$$\frac{S_T\cdot C_L^T}{S\cdot C_L}=\frac{L^T}{L}$$

$$\frac{C_L^T}{C_L}=\frac{L^T}{mg}\frac{S}{S_T}=0.01\frac{40}{5}=0.08$$

$$C_L=\frac{0.1}{0.1-0.08\cdot 5/8}=2$$

$$C_L^T=0.16$$

(ii)

$$\varepsilon=d\varepsilon/d\alpha\cdot C_L=0.1\cdot 2=0.2\,rad=11.46^\circ$$

$$\alpha_T=\alpha+\eta_T-\varepsilon=C_L/a+\eta_T-\varepsilon$$

$$\alpha_T=2/4.5\cdot 180/\pi-3-11.46=11^0$$

(iii)

$$C_L^T = a_1\alpha_T + a_2\eta + a_3\beta$$

$$\beta = 0 \Rightarrow \eta = (C_L^T - a_1\alpha_T)/a_2$$

$$\eta = (0.16 - 2.8\cdot 11\cdot \pi/180)/1.2 = -0.315 \text{ rad} = -18.03^0$$

(iv)

$$C_L^T = a_1\alpha_T + a_2\eta + a_3\beta$$

$$C_H = b_1\alpha_T + b_2\eta + b_3\beta, \quad C_H = 0$$

$$\eta = -(b_1\alpha_T + b_3\beta)/b_2$$

$$C_L^T = (a_1 - a_2 b_1/b_2)\alpha_T + (a_3 - a_2 b_3/b_2)\beta$$

$$\bar{a}_1 = a_1 - a_2 b_1/b_2 = 3.8, \quad \bar{a}_2 = a_3 - a_2 b_3/b_2 = 2.7$$

$$\beta = (C_L^T - \bar{a}_1\alpha_T)/\bar{a}_2 = (0.16 - 3.8\cdot 11\cdot \pi/180)/(-2.7) = 0.211 \text{ rad} = 12.09^0$$

$$\eta = (0.01\cdot 11\cdot \pi/180 - 0.03\cdot 0.211)/0.012 = -0.368 \text{ rad} = -21.06^0$$

3.3

(i)

$$C_L = -\frac{C_{m0}}{h - h_0 - \bar{V}_T C_L^T/C_L} = \frac{0.2}{0.1 - 5\cdot 0.08/8} = 4$$

$$C_L^T = 4\cdot 0.08 = 0.32$$

(ii)

$$\varepsilon = d\varepsilon/d\alpha \cdot C_L = 0.1\cdot 4 = 0.4\, rad = 22.9°$$

$$\alpha_T = \alpha + \eta_T - \varepsilon = 0.8\cdot C_L/a + \eta_T - \varepsilon$$

$$\alpha_T = 0.8\cdot 4/4.5\cdot 180/\pi - 3 - 22.9 = 14.84^0$$

(iii)

$$\eta = (C_L^T - a_1\alpha_T)/a_2$$

$$\eta = (0.32 - 2.8\cdot 14.84\cdot \pi/180)/1.2 = -0.338 \text{ rad} = -19.35^0$$

(iv)

$$\beta = (C_L^T - \bar{a}_1\alpha_T)/\bar{a}_2 = (0.32 - 3.8\cdot 14.84\cdot \pi/180)/(-2.7) = 0.246 \text{ rad} = 14.1^0$$

$$\eta = -(b_1\alpha_T + b_3\beta)/b_2$$

$$\eta = (0.01\cdot 14.84\cdot \pi/180 - 0.03\cdot 0.246)/0.012 = -0.399 \text{ rad} = -22.87^0$$

3.4

$$H_n' = h_0 - h + \bar{V}_T \frac{\bar{a}_1}{a}\left(1 - \frac{d\varepsilon}{d\alpha}\right)$$

$$\bar{V}_T = \frac{H_n' - (h_0 - h)}{\bar{a}_1 / a \cdot (1 - d\varepsilon / d\alpha)}$$

$$\bar{a}_1 = a_1 - a_2 \frac{b_1}{b_2} = 3 - \frac{1.8 \cdot 0.01}{-0.06} = 3.3$$

$$\bar{V}_T = \frac{0.12 + 0.18}{3.3/4.8 \cdot (1 - 0.47)} = 0.823$$

$$\bar{V}_T = \frac{S_T l_T}{S \bar{\bar{c}}}$$

$$S_T = \frac{\bar{V}_T S \bar{\bar{c}}}{l_T} = \frac{0.823 \cdot 10 \cdot 1.15}{10} = 1.23\,m^2$$

3.5

$$C_m = C_{m0} + (h - h_0) C_L - \bar{V}_T \left[\frac{\bar{a}_1}{a} \cdot \left(1 - \frac{d\varepsilon}{d\alpha}\right) C_L + \bar{a}_1 \eta + \bar{a}_3 \beta \right]$$

$$C_m = 0$$

$$\bar{a}_3 \beta_{trim} = \frac{C_{m0} + (h - h_0) C_L}{\bar{V}_T} - \frac{\bar{a}_1}{a}\left(1 - \frac{d\varepsilon}{d\alpha}\right) C_L - \bar{a}_1 \eta_T$$

$$\beta_{trim} = \left(\frac{C_{m0}}{\bar{V}_T \bar{a}_3} - \frac{\bar{a}_1 \eta_T}{\bar{a}_3} \right) + \frac{(h - h_0) C_L}{\bar{a}_3 \bar{V}_T} - \frac{\bar{a}_1}{a \bar{a}_3}\left(1 - \frac{d\varepsilon}{d\alpha}\right) C_L$$

$$\beta_{trim} = \left(\frac{C_{m0}}{\bar{V}_T \bar{a}_3} - \frac{\bar{a}_1 \eta_T}{\bar{a}_3} \right) - \frac{C_L}{\bar{V}_T \bar{a}_3} \left[(h - h_0) + \frac{\bar{V}_T \bar{a}_1}{a}\left(1 - \frac{d\varepsilon}{d\alpha}\right) \right]$$

$$\beta_{trim} = \left(\frac{C_{m0}}{\bar{V}_T \bar{a}_3} - \frac{\bar{a}_1 \eta_T}{\bar{a}_3} \right) - \frac{C_L}{\bar{V}_{Tbar} a_3} H'_n$$

3.6

$$\beta_{new} - \beta_{old} = -\frac{C_{L,new} - C_{L,old}}{\bar{V}_T a_3}$$

$$\bar{a}_3 = a_3 - \frac{a_2 b_3}{b_2} = 0.5 - \frac{2.3 \cdot (-0.08)}{-0.15} = 0.454$$

$$C_{L,old} = \frac{W}{0.5 \rho U_\infty^2 S} = \frac{3000}{0.5 \cdot 0.6 \cdot 180^2 \cdot 1.225} = 0.248$$

$$C_{L,new} = \frac{3000}{0.5 \cdot 0.74 \cdot 85^2 \cdot 1.225} = 0.916$$

$$\beta_{new} = \frac{-(0.916 - 0.278)\cdot 0.06}{0.55 \cdot 0.454} = 0.161 \text{ rad} = -9.2^0$$

3.7 (Pull-out manoeuvre, not in this year assessment, but we will experience it in the flight course)

$$\frac{\Delta\eta}{\eta}.\frac{\bar{V}_T a_2}{C_{L_1}} = h - h_0 - \bar{V}_T\left[\frac{a_1}{a}\left(1 - \frac{d\varepsilon}{d\alpha}\right) + \frac{a_1}{2\mu}\right]$$

$$\mu = \frac{m}{\rho S l_T} = \frac{65000}{1.225 \cdot 190 \cdot 14} = 19.95$$

$$C_{L_1} = \frac{mg}{½\rho V^2_e S} = \frac{65000 \cdot 9.8}{½ \cdot 1.225 \cdot 150^2 \cdot 190} = 0.2433$$

$$\Delta\eta = \frac{nC_{L_1}}{\bar{V}_T a_2}\left\{h - h_0 - \bar{V}_T\left[\frac{a_1}{a}\left(1 - \frac{d\varepsilon}{d\alpha}\right) + \frac{a_1}{2\mu}\right]\right\}$$

$$\Delta\eta = \frac{2.5 \cdot 0.2433}{0.49 \cdot 2}\left\{0.25 - 0.16 - 0.49\left[\frac{3.7}{4.5}(1 - 0.49) + \frac{3.7}{2 \cdot 19.95}\right]\right\} = -0.100 \text{ rad} = -5.72^0$$

E Avital